Notes from VSoN10 (July 2026)

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Day 1 (15th July 2026, Tuesday)

Neutrino physics - Introduction and first 50 years (Yuichi Oyama, KEK/J-PARC)

- KGF experiment and atmospheric neutrino detection.
- NC channel discovery in neutrino interaction.
- what is matter fermion?
- Why there is no matter consists of neutrinos similar to that of matter fermions?
  - *If we consider leptons as SU(2) doublets, it can be assumed neutrino interaction changes upper state to lowerstate before and after respectively.*
- Comparison between hadron-neuclei interaction $\sigma \approx 3 \times 10^{-26} cm^{-2}$ whereas, neutrino-neuclei interaction $\sigma \approx 1 \times 10^{-38} cm^{-2}$.
- *STUDY ABOUT SAVANA RIVER EXPERIEMT BY REINES AND COWAN*
  - This leads to discovery of anti-neutrinos.
  - **The basic principle:** two photons will come from $\bar{\nu}_e + p \to e^- + n$ and $e^- \to \gamma\gamma, n + Cd \to Cd^* \to Cd + \gamma$. i.e. two gammas detected withing 5 mircosec interval. Conicedence tecnique is use to detect the neutrino.
  - 3.0 ± 2.0 events per hour is recorded.
- Muon neutrino discovery by **AGS Experiment** by BNL.
  - *Physics Goal:*Same type of neutrinos (ν_e) as Savana river experiment or not.
  - *Spark chamber* is used to detect the neutrinos. Coincedence signal is used between two scintillation hits. Spark chamber gives long tracks for muons whereas for electron it shows a shower like character.
  - If neutrinos from muon decays are assumed to be electron type then total ~29 events expected whereas only 6 is obeserved.
  - Which gives an idea of different 2nd different of neutrinos.
  - *Is this 1st accelerator neutrino experiment??*
- *STUDY ABOUT KGF Experiment*
  - *Physics goal:* Is to observed atmospheric neutrinos.

- KGF have very small cosmic ray flux which helps to study neutrinos by removing cosmic background.
  - Plastic scintillator is used in this experiment.
  - PMTs are used to detect the photons.
  - Type I, Type II and Type III plastic scintillators.
  - All 3 events are catgorized into Type I experiment except from the atmospheric neutrinos.
  - Another South african mine experiments collieds with publication results from KGF.
  - Detection is same as the AGS experiment.
- *HAVE TO STUDY ABOUT HOMESTAKE EXPERIMENT!*

$$Cl + \nu_e \rightarrow Ar + e^-$$

- Davis Nobel prize at 2002.
- **Have to take into account that all these experiments was focus on CC interaction while detection. In Case of AGS experiment the production too.**
- Neutrino interaction via gauge bosons.
- Mass of W boson is 140 GeV which tells that they will exsists for very short time, from uncertainity principle $\Delta E.\Delta t \geq \hbar/2$.
- And time (t) is small hence, Range given by, $R \approx c\Delta t$ will be small.
- Cross section $\sigma \propto R^2$, the cross section of W boson are small, indicating hard to detect. **Is there any experiment to detect W boson?**
- **HAVE TO UNDERSTAND ABOUT THE MATRIX REPRESENTATION OF ELECTRO-WEAK INTERACTION**
  - W boson are off diagonal and Z boson is off diagonal. (*This may indiacte why NC is flavor independent!*)
- Similar to Z boson, photon is also diagonal matrix GWS model/Standard model.
- **HAVE TO STUDY ABOUT HIGGS PARTICLE AND SPONTANEOUS SYMMETRY BREAKING!**
- In 1973, NC Interactions were introduced. This is the experimental evidence of GWS model. NC interaction was discovered in **Gargamelle experiment** in CERN.
  - Gargamelle detector is a bubble chamber.
  - $\bar{\nu}_e e^- \rightarrow \bar{\nu}_e e^-$: **No muon neutrino produced**
  - $\bar{\nu}_\mu + N \rightarrow \bar{\nu}_\mu + hadrons$:**No muon neutrino produced**
  - **[Q] HOW THIS CAN BE CONCLUDED FROM THIS TWO ABOVE REACTION THAT THERE EXSISTS A NC CHANNEL!!**
- GWS was awarded because of the discovery of NC, which gave experimental evidence of GWS prediction of NC interaction.
- In 1983, W and Z boson discovery UA1/UA2 experiment in SppS, CERN.
- In 2012, Higgs particle discovery in LHC CERN.

Experimental neutrino physics concepts in a nutshell (Son Cao, IFIRSE)

Introduction to LBL:

- Why three flavor of neutrinos (/leptons)?
- What is a particle?
  - It is a excitation in of Quantum Fields.
- How particles have mass?
  - Particle have mass when they interact with Higgs particle. Number of interaction equivalent to mass. If interaction number is high mass of the particle will be high.
  - To interact with Higgs both chirality of the particles are needed.
  - That's why talking about the mass of neutrinos is a controversial topic as neutrino is left handed and antineutrino is right handed.
- **HAVE TO STUDY ABOUT THE NEUTRINO MASS GENERATION!!**
- **HAVE TO UNDERSTAND SUPERPOSITION OF MASS EIGENSTATES IN CONTEXT OF ROTATION.**
- $\theta_{13} \neq 0.0$ allows us to measure δ_{CP}.
- Some experiment like solar neutrino experiments consoders $\theta_{13} = 0.0$ approximation.
- Uncertainities are important in LBL experiments.
- Understanding of the datas are important. i.e. **You must know what you are doing!!**
- Why right handed neutrinos supposed to existis in low energy range?

Neutrino detection:

- **The first thing is we can not see neutrino, instead we look at the effects of neutrinos when interacts with matter such as nucleons and we look into the photons produced in the process with the help of the detector.**
- Things to note while detecting neutrinos:
  - Is this really neutrino?
  - Energy of the neutrinos?
  - Flavor of the neutrinos?
- Neutrinos produces very faint flash of lights in detectors. **What are the measure of it?**
- Neutrino nuclei interactions are very complicated.
- Material selection depends on the physics goal and economy of the experiment.
- *Simulation is imporant to understand the data.*

Histogram in HEP:

- Histogram is an important tool to visualize tha data.

- 2D and multidimensional histograms.
- **Signal :** Is the one which we want to study.
- **Background :** Is the anything else.
- In experimental physics, it is important to define what is the signal is.
  - Such example is ν_μ beam contains some fraction of ν_e.
- Ratio of signal to background is an imporant quantity to understand the data.
- Distinguishing between siganl and background is called **classification problem**.
  - Machine Learning is used to solve this problem.
- To make the selection of typically a likelihood of data to be signal or bkg is built which is called *particle identification (PID)*.

Hypothesis testing:

- H_0 : Null hypothesis what we want to reject.
- H_1 : what we want to examine.
- E.g. :
  - $H_0 \rightarrow$ Normal mass ordering.
  - $H_1 \rightarrow$ Inverted mass ordering.
- There are four possibilities:
  - H_0 true, H_1 false.
  - H_1 true, H_0 false.
  - Both true.
  - both false.
- Later two cases are difficult to deal with.
- There is errors related to hypothesis testing. Such as,
  - $H_0 : \theta_{13} = 0$
  - $H_1 : \theta_{13} \neq 0$.
- Error:
  - **Type I Error :** rejects H_0 even though it is true.
  - **Type II Error :** rejects H_1 even though it is true.
- **"Hypothesis can be ruled out, never be proven to be true." - Karl Popper**
- **HAVE TO LEARN ABOUT p-value**
- *T2K have good sensitivity to CP, NOvA has good sensitivity to CP but not good sensitivity to mass hierarchy.* - what does this mean?
- Parameter estimation using χ^2.

Systematics while reporitng the data:

- This is measure of uncertainities related to our experimnet.

- The needed to be included in the analysis.
- They must be included along with there correlations between them. **[Q] How to measure this correlation between them?**

Monte-Carlo simulation:

- Calculate pi value using MC simulation.
- Brownian motion MC.
- Toy model of virus transformation.

Handson traning:

- Tracking cosmic muon can help us to understand time dialation, Fermi coupling, test of Partiy-violation.

Day 2 (16th July 2026)

Particle and radiation detector 1/2 (Tsuyoshi Nakaya, Kyoto Univ., Hyper-K)

- X-rays and γ radiation.
- Detection of charged particle and neutral particle happens differently.
- **Books:**
    - Knoll
    - Leo
    - Introduction to expt. particle physics
    - detector for particle ration
- Difference between gamma ray and X-rays. **They are photons but depending production and energy gamma ray and X-ray are classified.**
    - X-ray comes from atomic process.
    - gamma photon comes from nuclear process.
- Nuclear physicist use half-life time whereas, particle physicist uses life-time (/Avg. life time)
- In photon detector we detect single photon. **MPPC can detect one photon. But what type of photon visible light, gamma photon or some thing else.**

Visble light:

$$E = h\nu$$

$$\lambda = \frac{1.240 \times 10^{-6}}{E}$$

where, λ in meters, E in eV

- So, we need two battery of 1.5 V to generate red lights.
- Sun temp. is 6000k, using Black body radiation we can determine the wavelenght from wein's displacement law. Then we can compute the Energy. Which will lie withing white range.
- Temp. of our body is 300k, photons from our body in long wavelegnth range.

Other radiation:

- α, β and cosmic ray.
  - Cosmic ray consist of muon (1 muon/100 cm^2/ec)
  - Cosmic proton (Which is hydrogen nucleus, the most abundant particle in universe) interact with atmopsheric Nitrogen and Oxygen to produce pion.
  - Pions decay to muon and neutrinos.

$$H^+ = P$$
$$P + N_2, O_2 \rightarrow \pi^{\pm,0} + X(hadrons)$$
$$\pi \rightarrow \mu + \nu_\mu$$

$$\tau_\mu = 20ns$$
$$\gamma c \tau_\mu = \gamma \times 3 \times 10^8 m/s \times 20 \times 10^9 s$$

Where, γ factor depends on energy of the proton energy and mass.

α particle:

- **[Q] Why α particle (He_2^4) is stable particle?**
  - s-orbit is most stable, 0^+ is more stable nuclear states.

X-rays and γ rays:

- Bremsstralung radition is source of γ ray, when relativistic charge particle propagates in electric field it emmits radition.
- Synchrotron radtion is also a source of gamma rays, when relativistic charge particle moves inside a magnetic field.

Energy loss of heavy charged particle:

- How much energy is deposited inside the matter by a charge particle.
- Bethe bloch can formula can be used to detemined the energy deposited in unit length of a material.

$$-\frac{dE}{dx} = 2\pi N_a r_e^2 m_e c^2 \rho \frac{Z}{A} \frac{z^2}{\beta^2} \left[\ln\left(\frac{2 m_e \gamma^2 v^2 W_{\max}}{I^2} \right) - 2\beta^2 - \delta - 2\frac{C}{Z} \right], \quad (2.27)$$

with

$$2\pi N_a r_e^2 m_e c^2 = 0.1535 \text{ MeVcm}^2/\text{g}$$

r_e: classical electron radius $= 2.817 \times 10^{-13}$ cm
m_e: electron mass
N_a: Avogadro's number $= 6.022 \times 10^{23}$ mol^{-1}
I: mean excitation potential
Z: atomic number of absorbing material
A: atomic weight of absorbing material
ρ: density of absorbing material
z: charge of incident particle in units of e
$\beta =$ v/c of the incident particle
$\gamma =$ $1/\sqrt{1-\beta^2}$
δ: density correction
C: shell correction
$W_{\max}$: maximum energy transfer in a single collision.

- **Have to do Bethe bloch formula derivation!!**
- **[Q] How neutron deposites energy in matter. It is a neutral particle.**
- **[Q] What is the momentum of a muon to beMinimun ionization of particle (MIP)?**

$$E = m\gamma$$
$$p = m\beta\gamma$$
$$\beta\gamma = \frac{p}{m}$$

$m_\mu = 100MeV/c^2 \equiv 0.1GeV/c^2$

$$\beta\gamma = 3$$
$$or, p/m = 3$$
$$or, p/(0.1Gev/c^2) = 3$$
$$or, p = 0.3GeV/c$$

  - **Why $\beta\gamma = 3$?**
    - From experimental plots.
  - Super-K have a diameter of 30m and for muon $-\frac{dE}{dx} = 2$ Mev/cm

$$E_{deposite} = -\frac{dE}{dx} \times L$$
$$or, E_{deposite} = 2MeV/cm \times 30 \times 1000$$

$$or E_{deposite} = 6000 MeV \equiv 6 GeV$$

Super-K can detect 6 GeV muons but in reality it can also detect higher energy muons.

- Even though Bethe-Bloch can be used for charged particle, but except electron beacuse during derivation of bethe-bloch we consider electron as target. Hence electron-electron collision will be not that much simple.
- Energy distribution of cosmic muon when pass through scintillator the distribution we get is called *Landau distribution*.
- While propagating charged particle losses energy due collsion as well as radiation such as Bremsstralung raditaion.

$$-\frac{dE}{dx} = \left(\frac{dE}{dx}\right)_{Rad.} + \left(\frac{dE}{dx}\right)_{Col.}$$

- **Radiation length:** When the energy of the radiation becomes 1/e times to that of the initial one.
- **Stopping power:** The distance when particle looses all the energy inside the medium.
- **Bragg's curve:** $\frac{dE}{dx}$ vs penitration length. This curve is used in medical treatment like cancer treatment.
- **Range:** The distance where the particle number is halved from the initial one.
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- **[Q] What is the range of pion with 100 MeV??**
- Defining the range of electron is not easy as it do not follow st. path due to different type of radition process. Extrapolation can be used to determine it's range.
- Interaction of e^+ is same while it is moving, but when it stops when anhilates and produce photons.
- **[Q] Why e^-e^+ decay via two channel one produced two photon and other three photon - why?**
  - $e^-e^+ \to \gamma\gamma$ *(Spin 0) : para-positronium*
  - $e^-e^+ \to \gamma\gamma\gamma$ *(Spin 1) : otho-positronium*
- **[Q] What is the lifetime of positronium?**

Standard model and neutrinos 1/2 (Nhung Dao, Phenikaa Univ.)

- Elementary particles are the fluctuation in quantum field.
- Electron and quarks are the smallest fundamentl particle till date.
- **[Q] How do we measure the size of the quarks?**
- Point like particle are those whose size is $< 10^{-18}$.
- **[Q] How can we say electric charge is quantozed when up quark have electric charge of $+\frac{2}{3}$**

- Stable particle those which never deacys, i.e. $\tau \rightarrow \infty$
- Decay width is the measure of the particle how do they decay into other particles.
- Three most unstable particle in Standard Model is : **Higgs, W, Z , top quark**.
- **Virtual particles :** Particles those donot follow $E^2 = m^2 + p^2$ relation.
- In QM, particle numbers are fixed but in QFT they are not.
- **[Q] Why in QFT we use Lagrangian formalism whereas in QM we prefer Hamiltonian formalism?**
- **[Q] Why Lagrangian have a mass dimension 4?**
- **Anhilition operator in QFT:** Anhilates one particle.
- **Creation operator in QFT:** Creates one particle.
- **[Q] What is gravitino?**
- **[Q] Even though neutron is fermion it do not have any anti-particle-why??**
    - No, neutrino have antiparticle called antineutron.
- **Pauli exclusion principle:** Two fermions can't be in same state.
    - That's fermionic field have anti commutation relation rather than commutation relation.
- Dirac field have mass dimension of 3/2. **WHY??**
- **[Q] Why Dirac field have four components?**
- **[Q] Why mass term do not respect/affect chirality, i.e. incase of mass term left chiral and right chiral part of Dirac field comes jumbled up - why??**
- So, to exhibit mass we need both left handed and right handed part of the particles, neutrinos do not have right handed part, so in standard model neutrinos are massless. To define mass we need to go beyond SM.
- Higgs paricle : real scalar field. (Scalar because spin 0 and real beacuse EM charge 0)
- gluons, photon, Z : real vector field (vector because spin 1 and real beacuse EM charge 0)
- W boson : Complex vector field (vector because spin 1 and real beacuse EM charge $\pm$1)
- quarks : Dirac field.
- leptons, neutrinos : weyl spinors.
- **Electromagnetic interaction treats left and right chirality equally.**
- **[Q] For weak interaction Lagrangian is of mass dimension of 6, but Lagrangian have to be of mass dimension of 4, what is happening?**
    - This can be take care of introduction of W bosons. **HOW!**
- In 1983, SPS (Super Proton Synchroton) CERN , W and Z bosons are discovered.
- **HAVE TO STUDY STRONG INTERACTION!**
- **TOPICS HAVE TO STUDY :**
    - Symmetry
    - Unitrary transformation
    - Abeliean Global symmentry
    - Abelian Gauge symmetry

    - QED
    - Non-Abeliean Gauge theory : Yang-Mills theory

Dark matter search with neutrino (Alba Domi, ECAP)

- **[Q] How can someone link with quantum gravity to neutrinos?**
- Quantum gravity can probe:
    - Black hole
    - Big bang
- *Quantum gravity is not renormalizable*.
- Success of standard model leads to used QED to string theory for unification of all forces.
    - **Fundamental principle:** Particles and gravity excitation of strings.
- General Relativity consider the curved manifold.
- No, experimental evidence of Quantum Gravity (QG).
- Plunk's scale is large. Which is difficult to reach in any current teristial experiments.
- Neutrino from different astrophysical source can solve this problem.
- **Quantum Coherence effect in neutrino oscillations**
- **Lorentz invariance violation**
- IceCube's neutrino energy threshold starts from TeV range and it can go to PeV range.
- **Have to study about Cherekkov detection!!!**
- **"When neutrino propagates it propagates in mass eigen state"-What does this mean?!**
- **Decoherence in neutrino oscillation:** If the osciillation are done in decohereance state that means oscillation is damped. which indicates that we will get different flavor at same detector other than the predicted by 3 flavor oscillation.
    - Lindblad equation: Markovian time evolution equation.
    - **Lorentz invariance violation!!-What's that?**
    - **What is lorentx viaolating fields?**
    - Effect of lattitude in oscillation probabilities in astrophyical and atmospheric neutrinos.

Day 3 (17th July 2026)

Particle and radiation detector 2/2 (Tsuyoshi Nakaya, Kyoto Univ.)

Interaction of photons:

- Photoelectric effect

- Crompton effect
- Pair production

Photoelectric effect:

- **[Q] Can this happen with free electron?**
  - Crompton scattering happens for free electron.
- **[Q] Does this process happen with the K-Shell e?**
- **[Q] Why does the probability becomes low at higher energy?**
- X-ray is emmitted after emition of photoelectron.
- Sometimes inseted of Xray emmission one L-shell electron is emmitted which is called Auger electron.

Compton effect:

- This happens for free electron but electron is never free it is bounded in metal. But what we assume is incident photon energy is very high than the binding eneergy of electron.
- **[Q] Using STR derive the Compton scattering formula.**
- **[Q] Calculate cross section using QED for Compton scattering.**
  - This formula is called Kelin-Nsihima formula.

Pair production:

-
  - **[Q] Calculate cross section using QED for Pair production.**

Electro magnetic shower:

- When high energy photon (GeV) is incidented squential pair production happens. which is difficult to calculate by hand. We need simulation like GEANT to simulate it.
- EM shower radiation legnth is characteristics of the material.
- **[Q] Under approximation how much length an e$^-$ can travel?**
  - For T2K $\nu_\mu \to \nu_e$ and the ν_e interact with neutron of water and produce electron by, $\nu_e + n \to e^- + p$.
  - The energy of the electron in approx 1 GeV
  - From yesterday we know, $-\frac{dE}{dx} \approx 2MeV/cm$
  - So electronn can travel 1GeV/2MeV = 5 m.
- For electron shower :

$$X_{max} \approx 4.5X_0$$

  - X_0 : for water is 36 cm.

Charge particle detection:

- Charge particle produces an electron which can be detected by collecting in anode.
- Scintillation uses observation of excited atoms to detect charged particle.
- There are different scintillator like organic and inorganic scintillator.
    - Mechanisms are little different. They work mostly on the band structure.

Detection of free electron after Excitation:

- Semi conductor detectors can be buse to dection of free electron after excitation.
- MPPC is such type of semiconductor detectors.
- These detectors have better resolution relative to other.

How to detect photons:

- Photon is interacting and electron is the measure of the photon.
- While detecting the photons advantage lies in higher stopping power of material and higher atmoic number.
    - **[Q] Why high atomic number needed?**
- Scintillation mechanism depends on several things such as:
    - Temp.
    - material and particles.
- **Photo Multiplier Tube (PMT)** : measures the photons
- **Semiconductor detectors**
    - In high energy physics these detector work in reverse bias.
    - Used in verted detector.
- **Superconducting detectors**
    - Have high resolution than semiconductors.
    - Used in Dark matter detection.
    - Economically costly.

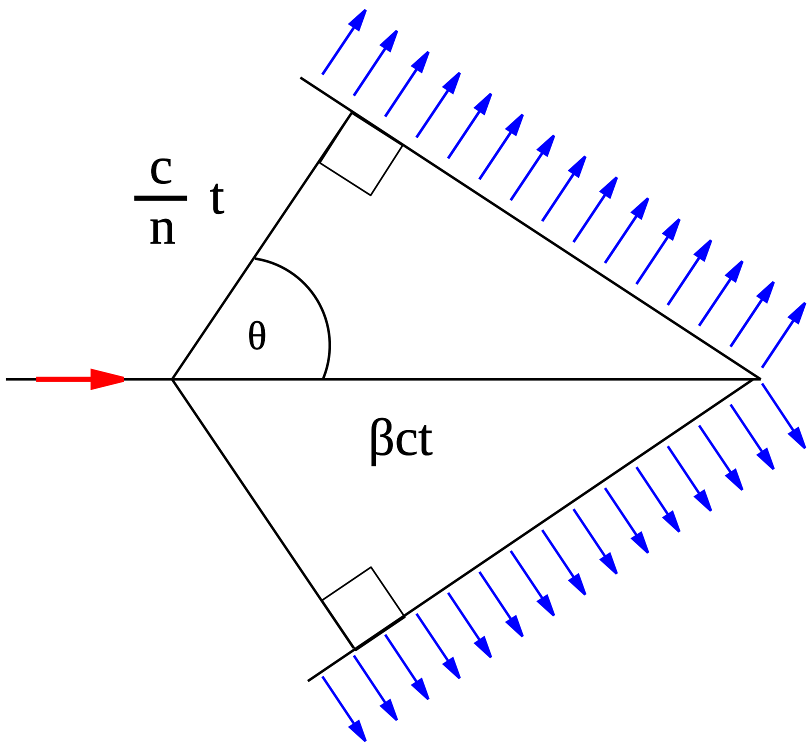
Different type of detectors:

- Cherenkov Detector
- Transition Raditaion Detector
- Time projection Chamber
- Nuclear Emulsion
- Nobel liquid detector

Cherenkov detector:

- Cherenkov radiation arises when charge particle in material moves faster than speed of light in the medium.
- This happens due to the polarization of the medium particle due to the charge particle.

$$cos\theta_c = \frac{ct/n}{\beta ct} = \frac{1}{\beta n}$$



Where, θ_c is the Cherenkov angle, $\beta = v/c$ and n is the refractive index. From this we can measure the angle of the chernkov ration from the vertex.

- **Disadvantages:** For Cherekov raditaion light yeild is very small.
- Cherenkov yeild is $\propto 1/\lambda^2$

Nuclear Emulsion detectors:

- It can be think as how photons are taken in 20th centuty.
- A thin film is exposed to radiation. Which produces an image which can be process to get a image.
- Cosmic muon can be detected using this detectors.
- Photon can be detected using this technique.

Liquid Ar TPC:

- Used in DUNE.

Standard model and neutrinos 2/2 (Nhung Dao, Phenikaa Univ.)

- The gauge group for SM: SU(3) $\otimes$ SU(2) $\otimes$ U(1).
  - SU(3)$_C$, C : for color charge. It has 8 generators with 8 gluons. Study of this strong force is called QCD.
  - SU(2)$_L$ $\otimes$ U(1)$_Y$:
    - This has 4 generators which leads to 4 gauge bosons W, Z and photon.
    - This symmetry must be broken to give mass to the particles.
- **HAVE TO STUDY SU(3)$_c$: STRONG INTERACTION**
- **HAVE TO STUDY SU(2)$_L$ $\otimes$ U(1)$_Y$ AND SYMMETRY BREAKING**
- Projection operator is $P_{L/R}$ and chiral operator is γ_5. **HAVE TO CLEAR THIS THING**
- Quark doublet can writen as $\begin{pmatrix} d_L \\ u_L \end{pmatrix}$ but have to make sure all the physics remains the same so charges (EM, Hypercharge) must be modified accordingly.
- **Number of independent element in CKM matrix:**

The Cabibbo-Kobayashi-Maskawa matrix 1973 (2)

$V_{\text{CKM}} = (V^{uL})^\dagger V^{dL}$: $N \times N$ unitary matrix, N: number of generations

- $N \times N$ unitary matrix is parameterized by N^2 independent parameter

  number of angles: $\frac{N(N-1)}{2}$, number of phases: $\frac{N(N+1)}{2}$

- One can remove phases by redefining fields

$$\hat{d}^i_L \to e^{\alpha^i_d} \hat{d}^i_L, \quad \hat{u}^i_L \to e^{\alpha^i_u} \hat{u}^i_L$$

$$\to \quad V^{ij}_{\text{CKM}} \quad \to \quad e^{i(\alpha^j_d - \alpha^i_u)} V^{ij}_{\text{CKM}}$$

  there are $2N-1$ independent relative phases which can be used to remove phases of the CKM matrix

- Therefore, number of remaining phases of CKM matrix

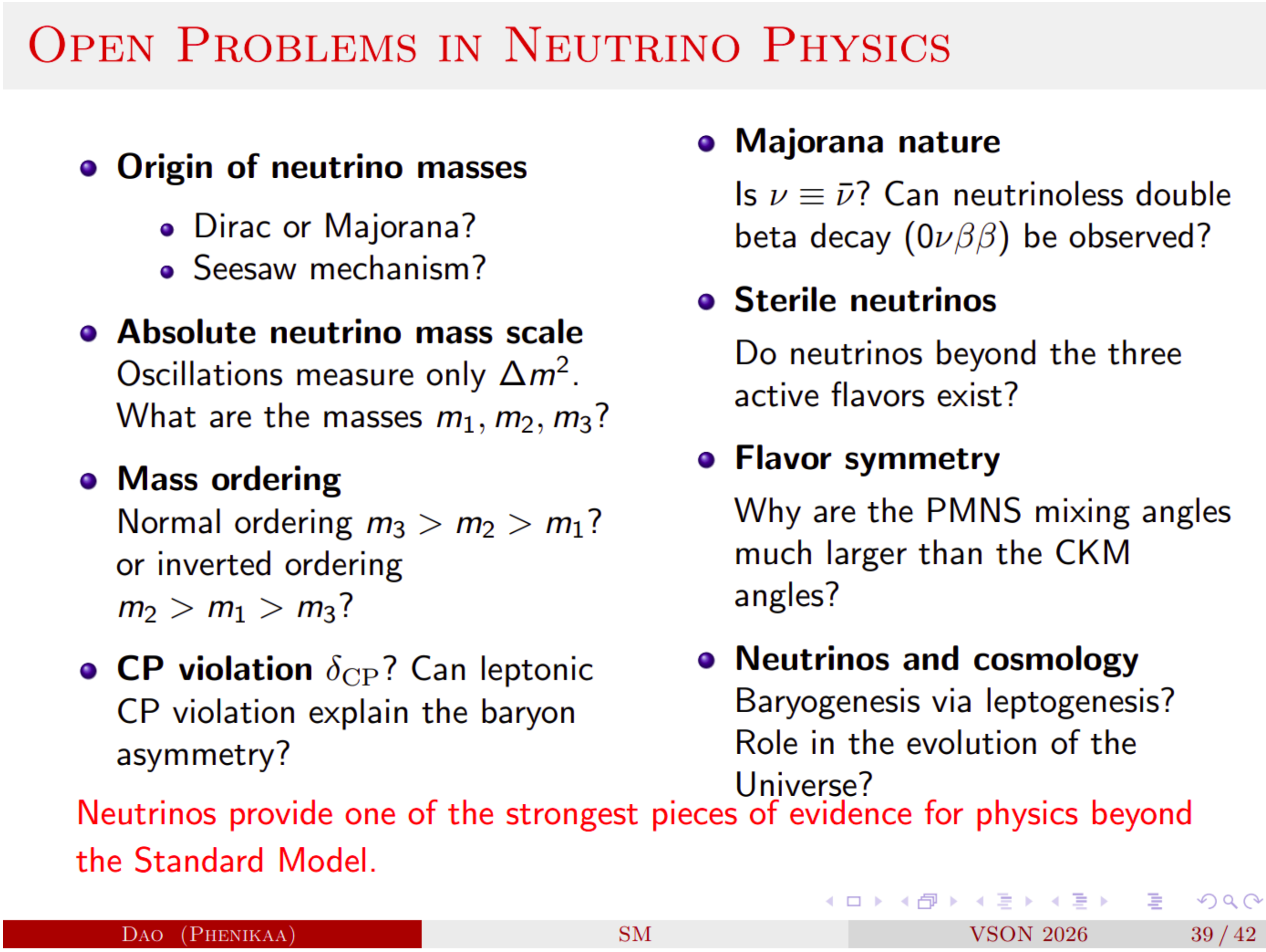
$$N_{\text{CP-phases}} = \frac{N(N+1)}{2} - (2N-1) = \frac{(N-1)(N-2)}{2}$$

N=1 or 2 then $N_{\text{CP-phases}} = 0 \to$ no CP violation, (Cabibbo)

N=3 then $N_{\text{CP-phases}} = 1 \to$ CP violation

$\to$ [Nobel price in 2008 for Kobayashi and Maskawa]

- This will be similar incase of PMNS too...
- There are still many questions on neutrinos such as:

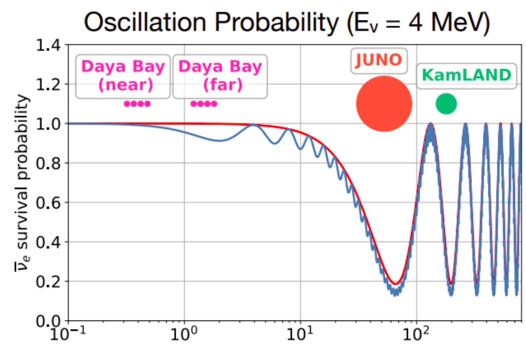


- **Have to study SEESAW mechanism!!**
- **The Majorana phase can be estimated from neutrino less double beta decay ($0\nu\beta\beta$). The event rate is proportional to the linear combination of both phases. So, probing them individually is difficult but we can have an estimate if there exists Majorana phase or not.**

Reactor neutrino experiments (Akira Takenaka, Sun Yat-sen Univ.)

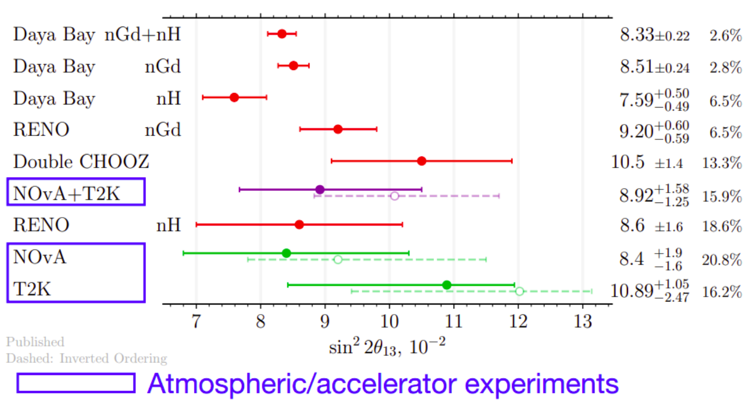
- **Have to study the Nobel literature by Renines for 1st reactor neurino flux measuremnt.**
- Scintillation photon are isotropic and Cherenkov photons have direction information. The number of photon produced in scintillation is higher than of the Cherenkov process.
- **In reactor neutrino experiment we can observe ν_μ, ν_τ as the antineutrino energy is low. So we observe antineutrino survival proability.**

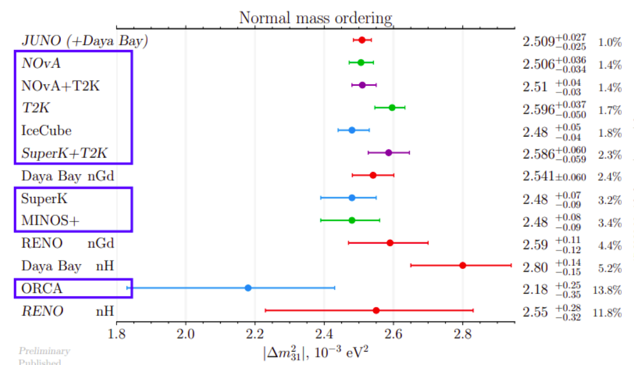
$$
\begin{aligned}
P(\bar{\nu}_e \to \bar{\nu}_\mu \text{ or } \bar{\nu}_\tau) &= \cos^4\theta_{13}\sin^2 2\theta_{12}\sin^2\left(\frac{\Delta m^2_{21}L}{4E}\right) \\
&\quad + \sin^2 2\theta_{13}\left[\cos^2\theta_{12}\sin^2\left(\frac{\Delta m^2_{31}L}{4E}\right) + \sin^2\theta_{12}\sin^2\left(\frac{\Delta m^2_{32}L}{4E}\right)\right] \\
P(\bar{\nu}_e \to \bar{\nu}_e) &= 1 - \underbrace{\cos^4\theta_{13}\sin^2 2\theta_{12}\sin^2\left(\frac{\Delta m^2_{21}L}{4E}\right)}_{\text{slow \& large}} \\
&\quad \underbrace{- \sin^2 2\theta_{13}\left[\cos^2\theta_{12}\sin^2\left(\frac{\Delta m^2_{31}L}{4E}\right) + \sin^2\theta_{12}\sin^2\left(\frac{\Delta m^2_{32}L}{4E}\right)\right]}_{\text{fast \& small}}
\end{aligned}
$$



- KamLAND reactor neutrino experiment can probe Δm^2_{21}, fast oscillation part for Δm^2_{31} is invisible for KamLAND.
- **Delayed coincidence technique** is used in Cowan and Reines, KamLAND experiment. **WHat is that!?**
- Daya Bay experiment can probe the fast oscillation i.e. $sin^2(2\theta_{13})$.
- Accelerator/atmospheric neutrino results aligns with each other for $sin^2(2\theta_{13})$ and $|\Delta m^2_{31}|$.

Status of θ13 & Δm²31





- **Why uncertainity in $sin^2(2\theta_{13})$ is high in accelerator experiment whereas for Δm^2_{31} uncertainity is high in reactor neutrino experiments ??** For JUNO Δm^2_{31} uncertainity is small. **WHY?**
  - This is purely due to statistical reason.
- **What is reactor neutrino anonmaly??**
- Reactor neutrino anomaly gives the motivation to study sterile neutrinos.
- JUNO uses same principle as KamLAND to detect neutrinos.

Day 4 (18th july 2026)

High energy neutrino astronomy and Supernova neutrino (Alba Domi, ECAP)

- **Neutrino astromoy**

High energy neutrino souces:

- Neutrino is an ideal candidate of messenger other than photons and protons.
- Photons interacts with CMB.
- Protons interacts with CMB and being electrically charged it deflected by magnetic field.
- Being Cross section small ($10^{-38}cm^2$ at 1 GeV) it can travle long distance.
- High energy neutrino produced by $P + matter \rightarrow \pi \rightarrow \nu + \mu$.
- Extragalactic and galactic neutrino sources.
- **AGN :** Active Galactic Nulcei, source of extragalactic neutrinos.
- **GRBs :** Gamma Ray Brusts, source of extragalactic neutrinos.
- Supernova reminant is source of galactic neutrino sources along with Magnestars.

Neutrino telescope:

- KM3Net, IceCube are the example of neutrino telescope.
- Detection principle include detection cherenkov photons produced in the medium.
- **HAVE TO STUDY CHERENKOV DETECTION!!**
- Photons are detected in PMTs.
- For water, Cherenkov angle $\theta_C = 42^\circ$
- ANTARES, Italy is Neutrino telescope.
- *ORCA-DU* : These are the detection units used in KM3Net.
- Optical background in water:
  - K40 decay.
  - Bioluminosence.
- Effective area measuremnt for estimating event rates in neutrino telescope.

Neutrino sources:

- **NGC 1068 :** Brightest neutrino source (AGN).
- **TXS 0506+056 :** First neutrino balazer (AGN).
- NGC donot have any jets towards the earth and TXS have jet in direction towards earths. These jets are the the source of the neutrinos.

- **Cosmic and cosmogenic origin** of neutrinos.
  - Cosmic Origin (Relic Background): Produced roughly 1 second after the Big Bang, these neutrinos are a primordial remnant of the early universe.
  - Cosmogenic Origin (Secondary Production): These are generated during the propagation of Ultra-High-Energy Cosmic Rays (UHECRs).
- **Neutrino radio detection**

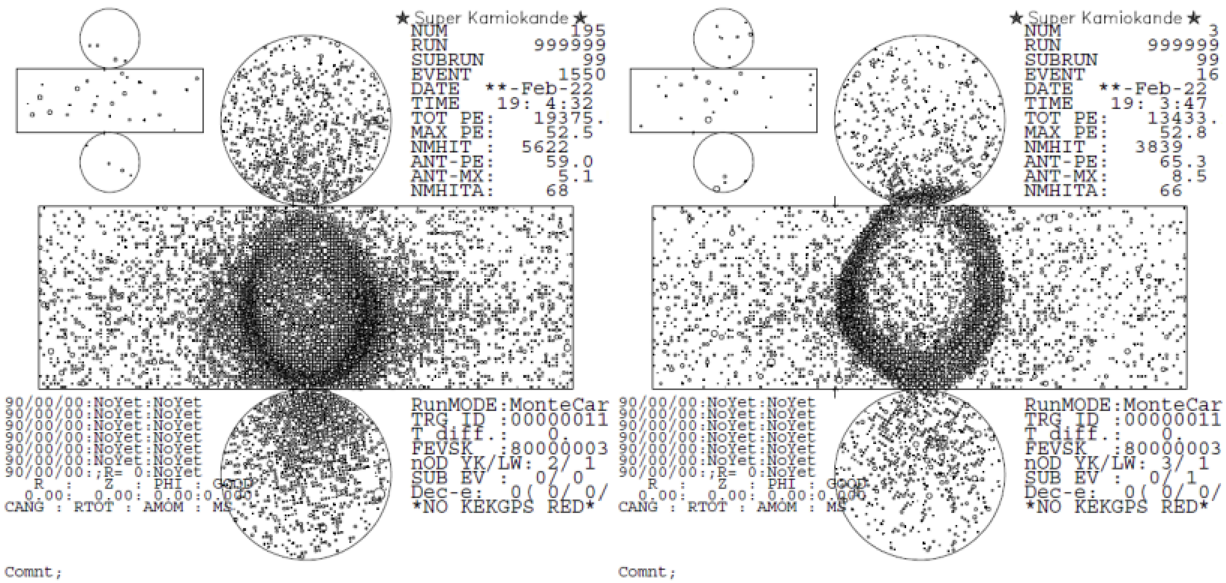
Supernova neutrinos:

- Supernovas are classified into 1a, 1b, 1c catagories.
- Supernova neutrino in MeV energy range.
- Kamiokande detected **SN 1987 A**.
- *Neutrino comes faster than photon from supernova because photon have a Brownian motion like behavior and also it interacts with matter. No a global aleart symstem is mainted called SuperNova Early Warning System (SNEWS) which tells all the neutrino and photon telescope to focus on the certain direction of the sky.*
- **DSNB** Diffused SuperNova Background.

Super-Kamiokande detector (Makoto Miura, ICRR Tokyo)

- KamiokaNDE was introduce to provide experimental evidence of GUTs via measuring proton lifetime.
- Background event of proton decay is neutrino event so they had to study neutrino.
- And they became neutrino experiment.
- At that time there were two big neutrino problems:
  - Solar neutrino problem
  - Atmospheric neutrino problem
- To investigate in detalis Kamiokande and IMB gourp collaborated to build Super-Kamiokande (SK).
- **Physics goals of SK:**
  - Atmospheric neutrino
  - Solar neutrinos
  - Supernova Neutrinos
  - Proton decay
  - Accelerator neutrinos (T2K)
- **[Q] Why Gd doped with pure water??**
  - SK used ultrapure water till 2018 which do not able to seperate between ν and $\bar{\nu}$ as SK can not identify charge but inclusion of Gd helps to distinguish between them by 8 MeV γ photon from neutron capture of Gd. **HOW!?**

- After 2022 when Gd dissolving was completed SK became SKGd.
- **And main physics target was changed to probe DSNB (Diffused SuperNova Background)**.
- SK uses cherenkov photon to detect neutrinos.
- SK tank contains 50 kt ultra pure water.
- **[Q] Calculate Cherenkov threshold for μ ($m_\mu = 106 MeV/c^2$) and π ($m_\pi = 140 MeV/c^2$).**
  - *Hint:* $P_{th} = m\beta\gamma = m\beta/\sqrt{(1-\beta^2)}$
- Outer shielding is added in SK out block cosmic rays.
- **SK Event display:**



  - Left one is for electron, fuzzy ring due to EM shower.
  - Right one is for muon, sharp ring no EM shower.
  - **[Q] Why elctron shows shower whereas muon shows long track?**

Software training: SK particle ID (Makoto Miura, ICRR Tokyo)

- **[Q] Why elctron shows shower whereas muon shows long track?**
  - Muon is 200 times heavier than electron and hard to make the showe.
  - Emits Cherenkov light until it stops.
  - Muons run longer distance than electron.

Day 5 (20th july 2026)

From Kamiokande to K2K 1/2 (Yuichi Oyama, KEK/J-PARC)

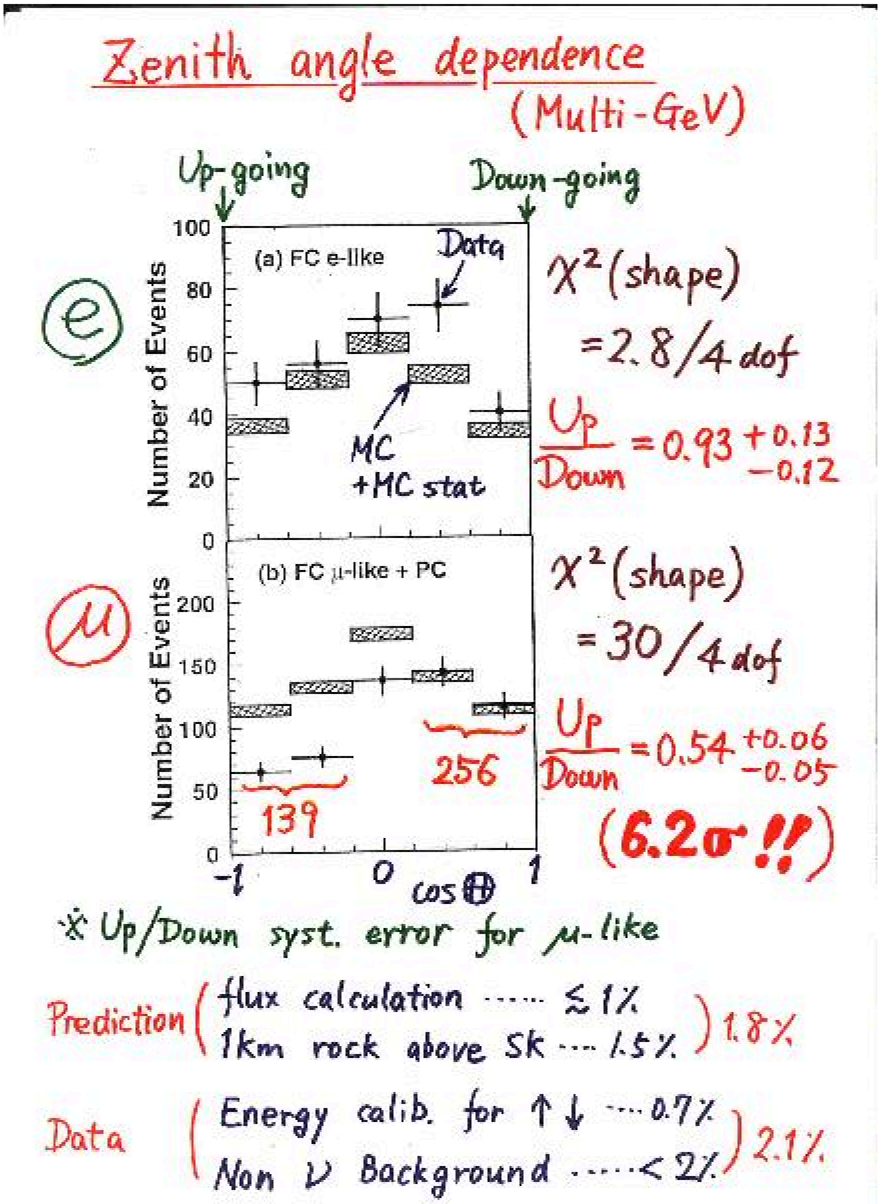
- GUT suggest quark's can be converted to leptons and the associated gauge boson is heavy M~ 10^{15} GeV.
- Just after bigbang this type of interaction were frequent.
- GUT is extension of GWS model.
- Neutrino decay is predicted by GUT.
- Which is the physics motivation of Kamiokande experiment.

Physics motivation of Kamiokande:

- Nucleaon lifetime is expected to be $10^{28} - 10^{32}$ years.
- Most hopeful model predicts life time to be 10^{28} years.
- Which leads to ≈ 6 nucleon decay per years.
- Back to back event is important for nucleon decay. **Why and how!!!??**
- No nucleon decay signal was found in Kamiokande after 2 years of observation.
- After that main physics goal changed to Kamioka Nucleon Decay Experiment to Kamioka Neutrino Detection Experiment.

Neutrino detection in Kamiokande:

- Kamiokande focuses on atmospheric and solar neutrino detection.
- Energy range of atmospheric neutrino is ~ 1 - 10 MeV for atmospheric neutrinos.
- Lower tigger rate allows to measure 8 MeV solar neutrinos too.
- Supernova **SN1987A** is detected by kamiokande in 1987 which is 160k light years away from our solar system.
- Results was confirmed by IMB experiment.
- They found 8 neutrino events with energies between 20 MeV and 40 MeV during 6 secs.
- In kamiokande paticle and antiparticles are not distinguished if not needed.
- Downaward ν_e, Downing going $\nu_m u$ and upgoing ν_e agrees with the expectation whereas upgoing ν_μ do not agrees with the expectation. Which is called atmospheric neutrino deficit.



- Atmosperhic neutrino deficit can be explained by $\nu_\mu \rightarrow \nu_\tau$ oscillation.

NUSEX experiment:

- NUSEX is a digital tracking calorimeter in Mont Blanc tunnel.
- The detector is made of iron plates.

Frejus experiment:

- Frejus experiment is located at France and Italy border.
- NUSEX and Frejus experiment denies Kamiokande results.

IMB-3 Experiment:

- It's results aligns with Kamiokande results.

Atmospheric neutrino oscillation:

- J/ψ **discovery of charm quark at 1974.**
- Atmospheric neutrino oscillation $\nu_\mu \rightarrow \nu_\tau$ is not belived by the spokes person of Kamiokande in 1992. This is due to the fact:
  - Capacity if e/μ paticle identification.
  - More statisctical data was needed.9
  - Uncertainity in atmospheric neutrino flux.
  - Negative results from other experiments like Frejus and NUSEX.

Neutrino interactions (Jake McKean, Kyoto Univ.)

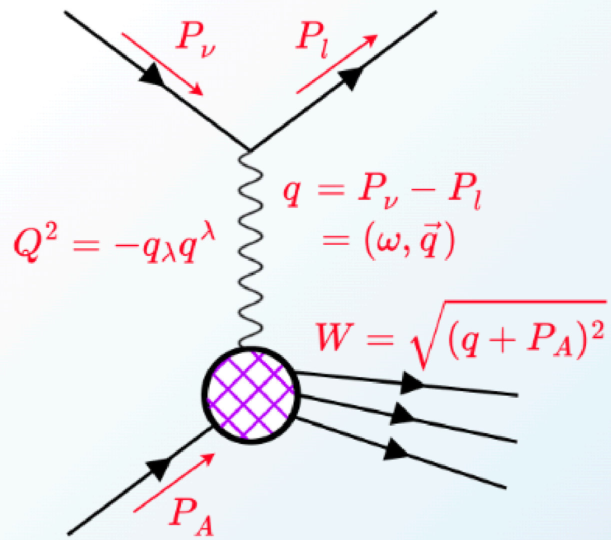
- Neutrino interaction estimation is important to understand the dynamics and physics how the particle interacts.
- Neutrino energy is not measured directly, it is measured throught the function of final state particle produced after interaction.
- Neutrino flux vs energy plot is important.
- **CC quasielastic scattering.**
- **NC quasielastic scattering.**
- For massless particle helicity and chirality are same.
- Weak interaction takes places for lefthanded particle.
- Which leads to the parity violation.
- Variation of cross section w.r.t different parameters tells the dependancy of the cross section with that parameter.
  - $\frac{d\sigma}{d\Omega}$: differential cross section wrt to solid angle.
  - $\frac{d\sigma}{dp_\mu}$: differential cross section wrt to muon momentum.
  - $\frac{d^2\sigma}{dp_\mu d\theta_\mu}$: double differential cross section wrt to momentum and angle.
- Depending upon the energy of the neutrinos sees different particles. At low energy neutrinos rarely interacts. At relatively higher energy neutrino see nucleons (most of the LBL opertaes in this range) and at very high energy neutrino intercts with quark and lots of particles are produced.

We will use these variables a lot, so we will define them here.

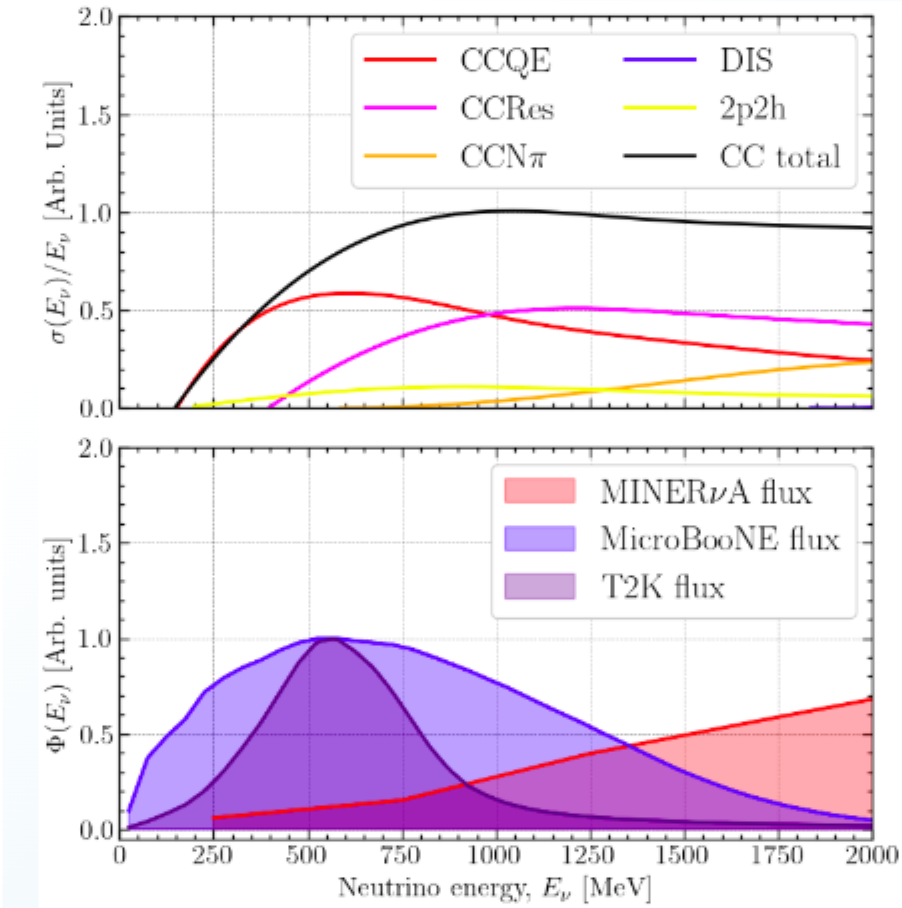
- **Q2**: tells you momentum scale of interaction
- **Omega:** tells you energy transferred to hadron system
- **W**: invariant mass of final hadronic system (W~1.2 GeV → resonance region (Delta 1232))

Large W and large Q2: lots of energy, now the neutrino resolves the constituent quarks → DIS!

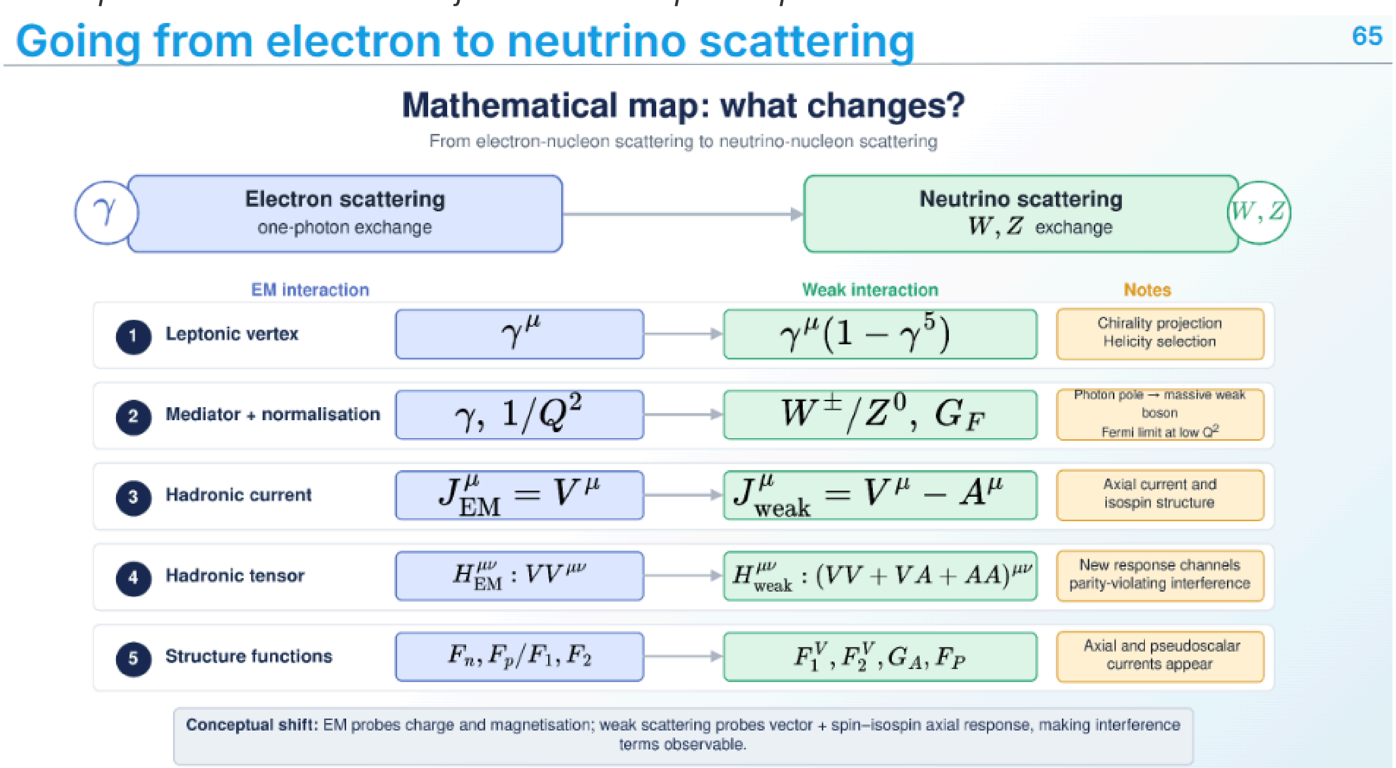
- **Inelasticity**: $y = \frac{q \cdot P_A}{P_\nu \cdot P_A} \overset{\text{rest frame}}{\Rightarrow} \frac{\omega}{E_\nu}$
- **Bjorken scaling**: $x = \frac{Q^2}{2P_A \cdot q}$



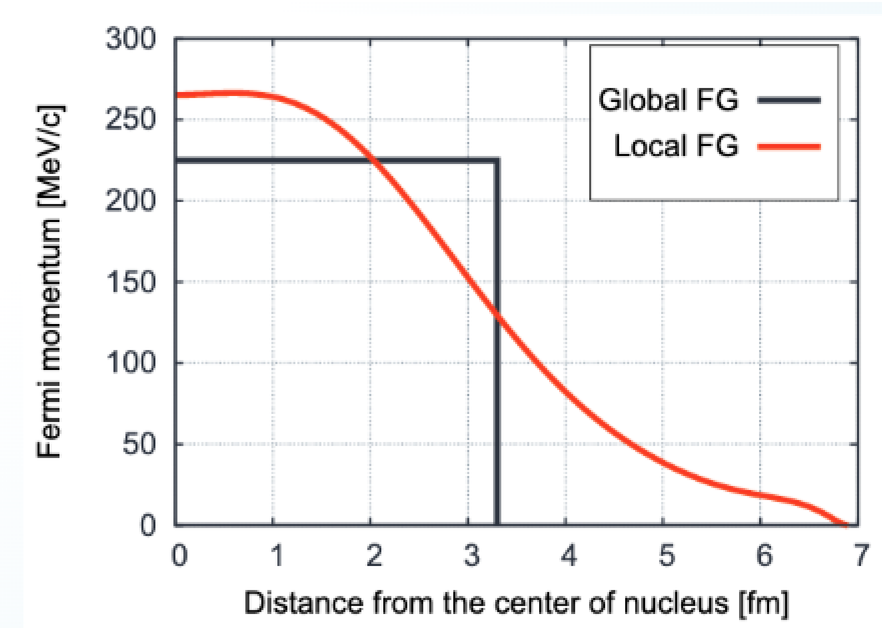
- **[Q] How to estimate the form factors???!!**
- Channels are named as $< current\ type >< interaction\ type >$, e.g.:
    - CCQE, NCQE, CCRes, CCDIS
- We have charged-current quasi elastic (CCQE) and neutral-current quasielastic (NCQE)
- At higher neutrino energies, we have CCRes & NCRes
- At even higher neutrino energies, there is deep inelastic scattering (DIS)
- Different interaction channels “turn on” at different neutrino energies
- This comes from kinematic constraints on the interaction, E.g.:
    - if there is not enough energy to produce a particle, then it won’t
- For different experimental fluxes, the dominating interaction channel is different, E.g.:
    - T2K is dominated by CCQE interactions.
    - but MINERvA includes more CCRes and DIS



- **Depending on the neutrino beam energy the beam contains flux from different processes.**
- In Deep Inelastic Scattering (DIS) which happens in higer energy range neutrinos interacts with quarks and in final state a large number of particles are produced.
- *Have to underastand e-nucleon interaction first then go for nucleon-nucleaon scattering. Which will help us a lot in this case we just need to map the lepton from electron to neutrinos.*



- To understand neutrino-neucleon interaction we are considering Final State Interaction (FSI) model. There are other things like:
  - Nuclear model, which nuclaear model we are considering : Here we consider **Fermi motion**.
  - And we also have to consider how neutrinos chooses which neucleaon to interact.
- For nuclear model we consider **Fermi gas model** which treats nucleus as fermi gas, which gives an estimate fermi energy, fermi momentum.
  - *Fermi momentum is an estimate of the binding energy of the neucleons inside the nucleous.*
  - *Neutrons are relatively highly bound than proton as proton are positively they repel each other.*
- Different nuclear models for interaction:
  - Relativistic Fermi Gas (RFG) model:
    - Global Fermi gas model: Uniform desnity (unrealistic)
    - Local fermi gas model : Not uniform density (More realistic)



  - Spectral function approach.
  - Mean-field theroy: It models nuclear force with meson exchange.
- In Final-state interactions we have to take account of several things, such as:
  - Elastic scattering
  - Nucleon rescattering
  - Charge exchange
  - Pion absorption
  - …
- **Impulse approximation :** Models the interaction on nucleus as an interaction with a single nucleon with other nucleons as spectators, then sum over available target nucleons. This turns many-body interaction into a sum of single-body interactions.
- This gives an brief idea how neutrino interacts with nucleons and nucleus.

Software training: NEUT (Ngoc Tran and Jake McKean)

MC sample:

MC method in event generator

NUET handson:

Whole slide

Day 6 (21st July 2026)

From Kamiokande to K2K 2/2 (Yuichi Oyama, KEK/J-PARC)

- **[Q] WHY TRACKING TYPE DETECTORS LIKE FREJUS AND NUSEX GAVE NEGATIVE RESULTS WHEREAS CHERENKOV DETECTOR GAVE POSTITIVE RESULTS WIH ATMOSPHERIC NEUTRINO?**
- **KEK Proton Synchrotron E261A**
- Beam test for particle identification. Such as for:
  - e beam.
  - μ beam.
- Super K running from 1996 - present, but it is modifed in different phases.
- Automatic reconstruction tool was introduced for super-kamiokande to classify between e and μ type events.
- **[Q] Why maximal mixing in atmospheric neutrino case??**
- *MACRO experiment (1989-2000)*:
  - Located in Italy.
- *Soudan-2 experiment (1989-2001)*:
  - Located in soudan mine, Minnesota, US.

Calculation of atm. neutrino flux:

- Calculation of atm. neutrino flux depends on:
  - Primary proton and heavy nuclei flux
  - Geomagnetic field.
  - Atmospheric tempareture.
- **The main takeaway is:** Event though the there were uncertainity in measureing atmospheric neutrino flux, the ν_e/ν_μ remains same.
- Large uncertainity while measuring atmospheric neutrino motivateed us to go for accelerator type neutrinos.

- There is pros and cons when we comapre atmospheric neutrio and accelerator neutrinos.
- *"Though there are many advantanges of accelerator neutrino over atmospheric neutrinos but first neutrino oscillation was discoverd using atmospheric neutrino experiment so atmospheric neutrino experiment is the winner in that way."* - **Oyama-San**
- **K2K experiment:**
    - KEK to Kamiokande.
    - Near detector is ~ 300m from target. It consist fine grain detector and a water cherenkov detector.
    - Far detector was Kamiokande water cherenkov detector.
    - Beam started 1999.
    - It collected 9.2×10^{19} POT equivalent events.
    - Time correlation is done using GPS. Beam window is 1.5 μs.
    - K2K results agrees with Super-K results.
    - This was the first experiment with accerator neutrino beam, to study atmospheric neutrino results.
- **MINOS experiment :**
    - ν_μ disappearance channel was studied.
    - 1st publication in 2006.
    - This independently confimed neutrino oscillations.
- **In 2015, Kajita-San (Super-K) and McDonald (SNO) was given nobel prize for $\nu_\mu \to \nu_\tau$ oscillation.**
- **[Q] Why donot we study ν_τ appearance channel?**
    - Detecting ν_τ is difficult because enregy of the neutrino must be high in multi GeV order, as mass of τ particle is ~ 1.78 GeV and from the expression below:

$$E_\nu > \frac{m_\tau^2 + 2m_\tau m_{nucleus}}{2m_{nucleus}} \approx 3.5\mathrm{GeV}$$

- L/E analysis helps us to probe neutrino decoherence and neutrino decay.
- After that arround 2000 there was several experiment which studied $\nu_e \to \nu_\mu$.
- This leads to study of three flavor oscillations.
- **NOTE :** *In cherenkov type detector ν_τ creates many number of ring instead of single ring expected in e and μ type events.*

Neutrino phenomenology 1/2 (Sanjib Agarwalla, IOP Bhubaneswar and UW-Madison, online)

- **Active neutrinos :** The neutrinos which interacts via weak interaction (through W and Z bosons).
- **Sterile neutrinos :** Do not interact via weak interaction.

- **[Q] How do we predict spin of neutrino?**
  - Beta decay :

$$n \rightarrow p + e^- + \nu$$

    - spin of n, p, e : $\frac{1}{2}$
    - So neutrino have to be a spin 1/2 to maintain the angular momentum conservation.
    - **[Q] Can we plot the beta decay spectrum from theory and compare with experiment?**
- LEP experiment predicted three flavor of neutrinos.

$$e^+e^- \rightarrow Z \rightarrow \nu\bar{\nu}$$

- **Ref :** LEP Phys. Rept. 2005
- **[Q] What is the minimum energy of the incoming neutrino required to create corresponding leptons??**
- DOUNT experiment and ν_τ discovery.
  - **Ref:** Observation of Tau Neutrino by DONUT collaboration at Fermilab
- **IceCube, KM3Net, ORCA experimt to study astrophysicsal neutrinos.**
  - KM3Net have detected highest energy neutrino event of energy 220 PeV.
    - **Ref :**The Highest-Energy Neutrino Event Constrains Dark Matter-Neutrino Interactions
- **PTOLEMY experiment to study Relic neutrinos**
- **[Q] What is Glawshow resonance?**
  - At neutrino energy ~ 6.05 PeV, a sudden peak in cross section is obsereved. Due to production of W boson in s-channel in antineutrino-e scattering process.
  - IceCube in 2021, showed that they have seen 1 Glashow resonance like event in their detectors.
  - **Ref :**Detection of a particle shower at the Glashow resonance with IceCube
  - This is one type of Brigth-Wigner resonace.
  - **[Q] What is Brigth-Wigner resonace??**
- **[Q] Calculate threshold energy for different proccess??**
-